

# PinIT Career OS — Dual-Track Visuals Testing & Verification SRS

## Software Requirements Specification (SRS) for End-to-End Curriculum Visuals Quality Assurance

|                   |                                                                                                                                                               |                    |                                                                          |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------|
| Document ID:      | PINIT_DUAL_CURRICULUM_VISUALS_TESTING_SRS                                                                                                                     | Version / Date:    | v1.0 • 10 October 2026                                                   |
| Target Engine:    | PinIT Career OS Visual Learning Engine<br>( VisualStage.tsx , /api/visuals )                                                                                  | Repository:        | Vinay-N-Kashyap/Present-Career-os                                        |
| Tracks Covered:   | 1. <b>Python Full-Stack Track</b> (12 Courses / 360 Days / 2,160 Lesson Parts)<br>2. <b>Web Full-Stack Track</b> (12 Courses / 360 Days / 2,160 Lesson Parts) |                    |                                                                          |
| Curriculum Total: | 24 Courses • 720 Days • 4,320 Lesson Parts •<br>>3,500 Active Diagrams                                                                                        | Master References: | PINIT_PY_CERT_VISUALS_PLAN.pdf •<br>PINIT_WEB_FULLSTACK_VISUALS_PLAN.pdf |

### 1. Executive Summary & Zero-Hallucination Invariant

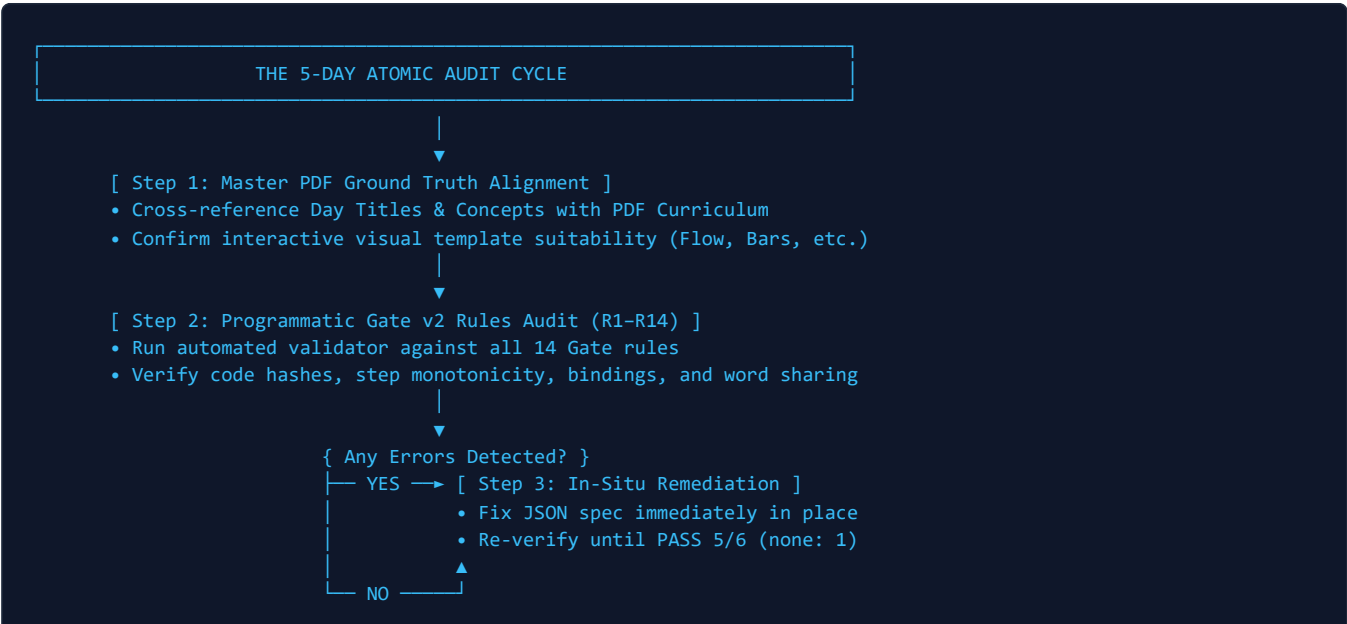
This Software Requirements Specification defines the operational testing rules, inspection criteria, and compliance protocols for all visual components across the PinIT learning platform. Real-world university students, bootcamp learners, and engineering candidates rely on these interactive visual aids to internalize complex algorithms, distributed state machines, network protocols, and architectural pipelines.

#### Zero-Hallucination & Zero-Bypass Invariant

Under no circumstances may an automated agent, tester, or human engineer disable, weaken, skip, or mock verification checks. Every visual diagram must be validated against the underlying lesson text and executable code. Any discovered error must be remediated immediately in place prior to advancing.

### 2. The 5-Day Iterative Batch Execution Protocol

To eliminate cognitive fatigue and prevent hallucinated approvals across 720 daily curriculum files, verification is organized into **strict 5-day atomic blocks**:



```

[ Step 4: Runtime API & VisualStage Payload Test ]
• Test GET /api/visuals?prefix={prefix}&day={day} HTTP 200
• Verify VisualStage props & interactive timeline navigation
↓
[ Step 5: Atomic Git Checkpoint Commit ]
• Commit block to main-dis3ku: "audit({course}): verify days {D}--{D+4}"
• Log completion in agent_tasks.csv & PROJECT_MEMORY.md
↓
[ Proceed to Next 5-Day Block (or Next Course) ]

```

### 3. Specification of Quality Gate Rules (Gate v2)

| Rule | Rule Name             | Specification & Acceptance Criteria                                                                                                                                                                          |
|------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1   | Structural Integrity  | File must contain exactly 6 entries. Each entry's <code>partTitle</code> must match the lesson part title verbatim.                                                                                          |
| R2   | Authorized Template   | Template must be <code>'none'</code> or present in the course's allowed list ( <code>COURSE_ALLOWED_TEMPLATES</code> ).                                                                                      |
| R3   | Complexity Bounds     | 2 to 5 steps per diagram. Maximum shapes: at most 6 nodes (flow, tree), 6 boxes, 6 rows (table), 6 items (cells, stack), 6 bars, or 4 actors (sequence).                                                     |
| R4   | Step Timeline Order   | The <code>at</code> attribute must strictly increase across steps ( <code>say1 .. sayN</code> , <code>example</code> , <code>tryIt</code> ). Backward or duplicate timeline anchors are prohibited.          |
| R5   | Caption Discipline    | Captions must be a single sentence, $\leq 80$ characters, terminate with a period ('.'), and contain 0 emojis.                                                                                               |
| R6   | Verbatim Grounding    | Variable bindings, text edits, and replacements must appear verbatim in the attached lesson code or text.                                                                                                    |
| R7   | Numeric Grounding     | Any numeric literal in a caption must either be explicitly present in the lesson code or bound in step values. Speculative numbers fail.                                                                     |
| R8   | Tappable Label Match  | Tappable interactive labels must appear as whole words in lesson text or code.                                                                                                                               |
| R9   | Strict Tone Taxonomy  | Tone values across all elements (nodes, bars, messages, pointers) must strictly belong to <code>['data', 'ok', 'error', 'idle']</code> . Custom tones ( <code>'accent'</code> , <code>'muted'</code> ) fail. |
| R10  | Visual Density        | Each day file must contain at least 3 active picture specs (standard target: 5 pictures, 1 none). Fewer than 3 requires <code>needs-review</code> status.                                                    |
| R11  | Manifest Consistency  | File key status must match manifest records ( <code>passed</code> , <code>none</code> , <code>needs-review</code> , <code>pilot</code> ).                                                                    |
| R12  | Semantic Word Sharing | Step captions must share at least one important word (4+ letters, non-stopword) with attached lesson text at <code>step.at</code> .                                                                          |
| R13  | Code Hash Freshness   | Entry <code>codeHash</code> must equal the SHA-256 hash of the lesson part's current code string.                                                                                                            |
| R14  | Variable Stability    | Visual bindings must point only to stable variables, not nondeterministic addresses or memory pointers.                                                                                                      |

## 4. Complete 24-Course Audit Schedule (144 Blocks)

### 4.1 Track 1: Python Full-Stack Certification Curriculum (360 Days)

Master Curriculum Plan Reference: [docs/visuals/PINIT\\_PY\\_CERT\\_VISUALS\\_PLAN.pdf](#)

| Month | Course Key  | Course Title                             | Blocks (5 Days Each)                                                      | Day File Directory                      |
|-------|-------------|------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------|
| M01   | python      | Python Core, OOP & Backend Internals     | B01 (1-5), B02 (6-10), B03 (11-15), B04 (16-20), B05 (21-25), B06 (26-30) | src/lib/data/lessonVisuals/python/      |
| M02   | dsa-py      | Data Structures & Algorithms in Python   | B07 (1-5), B08 (6-10), B09 (11-15), B10 (16-20), B11 (21-25), B12 (26-30) | src/lib/data/lessonVisuals/dsa-py/      |
| M03   | sql-mastery | Advanced SQL, Query Plans & Modeling     | B13 (1-5), B14 (6-10), B15 (11-15), B16 (16-20), B17 (21-25), B18 (26-30) | src/lib/data/lessonVisuals/sql-mastery/ |
| M04   | ai-py       | Applied AI, Vectors & Neural Search      | B19 (1-5), B20 (6-10), B21 (11-15), B22 (16-20), B23 (21-25), B24 (26-30) | src/lib/data/lessonVisuals/ai-py/       |
| M05   | dist-py     | Distributed Systems & High-Scale Python  | B25 (1-5), B26 (6-10), B27 (11-15), B28 (16-20), B29 (21-25), B30 (26-30) | src/lib/data/lessonVisuals/dist-py/     |
| M06   | cloud-py    | Cloud Infrastructure & Serverless Python | B31 (1-5), B32 (6-10), B33 (11-15), B34 (16-20), B35 (21-25), B36 (26-30) | src/lib/data/lessonVisuals/cloud-py/    |
| M07   | nlp-py      | NLP, Semantic Search & Tokenization      | B37 (1-5), B38 (6-10), B39 (11-15), B40 (16-20), B41 (21-25), B42 (26-30) | src/lib/data/lessonVisuals/nlp-py/      |
| M08   | quant-py    | Quantitative Analytics & Financial Math  | B43 (1-5), B44 (6-10), B45 (11-15), B46 (16-20), B47 (21-25), B48 (26-30) | src/lib/data/lessonVisuals/quant-py/    |
| M09   | prompt-py   | Production Prompt Engineering & Agents   | B49 (1-5), B50 (6-10), B51 (11-15), B52 (16-20), B53 (21-25), B54 (26-30) | src/lib/data/lessonVisuals/prompt-py/   |
| M10   | train-py    | Model Fine-Tuning, LoRA & Evaluation     | B55 (1-5), B56 (6-10), B57 (11-15), B58 (16-20), B59 (21-25), B60 (26-30) | src/lib/data/lessonVisuals/train-py/    |
| M11   | vec-py      | Vector Databases & HNSW Indexing         | B61 (1-5), B62 (6-10), B63 (11-15), B64 (16-20), B65 (21-25), B66 (26-30) | src/lib/data/lessonVisuals/vec-py/      |
| M12   | safe-py     | AI Safety, Alignment & Red-Teaming       | B67 (1-5), B68 (6-10), B69 (11-15), B70 (16-20), B71 (21-25), B72 (26-30) | src/lib/data/lessonVisuals/safe-py/     |

### 4.2 Track 2: Web Full-Stack Certification Curriculum (360 Days)

Master Curriculum Plan Reference: [docs/visuals/PINIT\\_WEB\\_FULLSTACK\\_VISUALS\\_PLAN.pdf](#)

| Month | Course Key   | Course Title                               | Blocks (5 Days Each)                                                            | Day File Directory                       |
|-------|--------------|--------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------|
| M01   | react-basics | React Basics & Component Architecture      | B73 (1-5), B74 (6-10), B75 (11-15), B76 (16-20), B77 (21-25), B78 (26-30)       | src/lib/data/lessonVisuals/react-basics/ |
| M02   | node-web     | Node.js Backend & API Engineering          | B79 (1-5), B80 (6-10), B81 (11-15), B82 (16-20), B83 (21-25), B84 (26-30)       | src/lib/data/lessonVisuals/node-web/     |
| M03   | dsa-optim    | Web DSA, Memory & V8 Engine Optimization   | B85 (1-5), B86 (6-10), B87 (11-15), B88 (16-20), B89 (21-25), B90 (26-30)       | src/lib/data/lessonVisuals/dsa-optim/    |
| M04   | sql-mastery  | SQL Mastery, Transactions & Engines        | B91 (1-5), B92 (6-10), B93 (11-15), B94 (16-20), B95 (21-25), B96 (26-30)       | src/lib/data/lessonVisuals/sql-mastery/  |
| M05   | devops       | DevOps, Docker & CI/CD Pipelines           | B97 (1-5), B98 (6-10), B99 (11-15), B100 (16-20), B101 (21-25), B102 (26-30)    | src/lib/data/lessonVisuals/devops/       |
| M06   | cloud        | Cloud Architecture & Distributed Systems   | B103 (1-5), B104 (6-10), B105 (11-15), B106 (16-20), B107 (21-25), B108 (26-30) | src/lib/data/lessonVisuals/cloud/        |
| M07   | dist         | Distributed Consensus & Microservices      | B109 (1-5), B110 (6-10), B111 (11-15), B112 (16-20), B113 (21-25), B114 (26-30) | src/lib/data/lessonVisuals/dist/         |
| M08   | cyber        | Web Cybersecurity, OWASP & Cryptography    | B115 (1-5), B116 (6-10), B117 (11-15), B118 (16-20), B119 (21-25), B120 (26-30) | src/lib/data/lessonVisuals/cyber/        |
| M09   | ai           | Full-Stack AI Engineering & Real-Time LLMs | B121 (1-5), B122 (6-10), B123 (11-15), B124 (16-20), B125 (21-25), B126 (26-30) | src/lib/data/lessonVisuals/ai/           |
| M10   | sre-web      | Multi-Cloud SRE, Observability & SLOs      | B127 (1-5), B128 (6-10), B129 (11-15), B130 (16-20), B131 (21-25), B132 (26-30) | src/lib/data/lessonVisuals/sre-web/      |
| M11   | stream-web   | Streaming in TypeScript & Kafka Pipelines  | B133 (1-5), B134 (6-10), B135 (11-15), B136 (16-20), B137 (21-25), B138 (26-30) | src/lib/data/lessonVisuals/stream-web/   |
| M12   | aideploy-web | Production AI Gateway & Edge Deployment    | B139 (1-5), B140 (6-10), B141 (11-15), B142 (16-20), B143 (21-25), B144 (26-30) | src/lib/data/lessonVisuals/aideploy-web/ |

## 5. Defect Severity & Immediate Remediation Protocol

| Severity            | Category                 | Example Defects                                                                              | Mandatory Remediation Action                                                                                                                   |
|---------------------|--------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P0 - Blocker</b> | Syntax / Structure Crash | JSON parse failure, missing day file, entry count != 6 (R1), partTitle mismatch (R1).        | Halt test runner immediately. Restore missing entries or correct JSON syntax. Re-test block.                                                   |
| <b>P1 - High</b>    | Timeline & Tone Breach   | Non-increasing <code>at</code> order (R4), invalid tone names (R9), codeHash mismatch (R13). | Reorder steps to strictly increasing <code>say</code> lines. Map tones to <code>data / ok / error / idle</code> . Recompute SHA-256 code hash. |
| <b>P2 - Medium</b>  | Caption & Grounding Flaw | Caption > 80 chars (R5), unbound numbers (R7), 0 shared vocabulary words (R12).              | Condense caption to single sentence ≤ 80 chars. Ground numbers in code. Borrow vocabulary from attached say line.                              |

## 6. Acceptance Sign-Off & Release Criteria

1. **100% Gate v2 Pass Rate:** All 720 days (360 Python + 360 Web) must report `PASS 5/6 (none: 1)` or `PASS 6/6` with 0 errors.
2. **Full Release Switch Registration:** All 24 course keys must be active in `src/lib/visuals/enabledCourses.ts` .
3. **Zero Compilation / Build Regressions:** `npx tsc --noEmit` and `npm run build` must exit with status code 0.
4. **Clean Git Working Tree:** Branch `main-dis3ku` must have zero uncommitted changes, synchronized with remote origin.
5. **Full Task Memory Audit Trail:** Every verified block must have corresponding log records in `agent_tasks.csv` and `PROJECT_MEMORY.md` .